

Codebook for www.conflictforecast.org (8 March 2023)

This codebook provides some technical details on the provided datasets, forecast methods and some guidance on the eight models we provide.

Country level predictions

We predict three types of conflict measures:

1. **An outbreak of any violence: a country goes from no fatalities to more fatalities in a month.**
2. **An outbreak of armed conflict: a country goes from having less than 0 to 0.5 fatalities per 1 million inhabitants in a month to more.**
3. **Intensity of armed conflict. We predict the $\ln$ of fatalities per 1000 population + 1 but report our forecast in terms of fatalities.**

Using two sets of variables:

1. Text model: Using 15 topic shares and number of tokens (words)
2. Best model: Using 15 topic shares and information about past violence

Over two horizons:

1. 3 months ahead: Aggregate conflict measure over the entire next 3 months
2. 12 months ahead: Aggregate conflict measure over the entire next 12 months

Our forecast relies on several variables that capture the conflict history and intensity of a country, the 15 topics we generate with the topic model and a count of the number of tokens. We combine these in different random forest models to provide our forecasts. Each random forest model uses a different set of hyper parameters which are calibrated using the forecast performance in the period 2010 to 2021.

The twelve models we provide at the country level are:

- Armed conflict, 12 months ahead, best model
- Armed conflict, 12 months ahead, text model
- Armed conflict, 3 months ahead, best model
- Armed conflict, 3 months ahead, text model
- Any violence, 12 months ahead, best model
- Any violence, 12 months ahead, text model
- Any violence, 3 months ahead, best model
- Any violence, 3 months ahead, text model
- Violence intensity, 12 months ahead, best model
- Violence intensity, 12 months ahead, text model
- Violence intensity, 3 months ahead, best model
- Violence intensity, 3 months ahead, text model

We provide these risk forecasts even for countries which are in conflict as they might transition into peace and back to conflict.

At the country level the map shows a mix of risks taken from the armed conflict 12 months ahead forecast and the violence intensity 12 months ahead forecast. In both cases we take the best model.

Note that when we predict 3 or 12 months ahead, then we are not predicting each of the next months separately. We are predicting the aggregate over the entire interval of the next 3 or 12 months.

Subnational predictions

At the subnational level we predict the probability of any conflict death at the 55km x 55km grid cell level within the following 12 months using a model using a random forest. The predictors include:

- topic shares at the grid cell level
- Information about past violence in the grid cell
- Information about violence in neighboring grid cells
- Information about at the country level
- Geographic features of the grid cell (mountainous terrain, distance to capital, population, etc)

Details on country level the datasets

The datasets provided for download are named according to the variable that is forecasted (armedconf3 contains the three months ahead forecast of armed conflict, for example).

The variable we provide are:

- country isocode
- year
- month
- text_model: the forecast probability of the text model that internal political violence (armedconf/anyviolence) breaks out within the next 3 or 12 months
- best_model: the forecast probability of the best model that internal political violence (armedconf/anyviolence) breaks out within the next 3 or 12 months
- topic 0: share of topic 0 in the text that country/month ("middle east")
- topic 1: share of topic 1 in the text that country/month ("civilian life")
- topic 2: share of topic 2 in the text that country/month ("competition and sports")
- topic 3 share in the text that country/month ("international cooperation")
- topic 4 share in the text that country/month ("military and diplomacy")
- topic 5 share in the text that country/month ("military conflict ")
- topic 6 share in the text that country/month ("Chinese politics")
- topic 7 share in the text that country/month ("politics")
- topic 8 share in the text that country/month ("power and negotiation")
- topic 9 share in the text that country/month ("foreign policy")
- topic 10 share in the text that country/month ("military technology")
- topic 11 share in the text that country/month ("national development")
- topic 12 share in the text that country/month ("energy and industry")
- topic 13 share in the text that country/month ("judiciary and abuses")
- topic 14 share in the text that country/month ("health and emergencies")

- armedconf: our definition of armed conflict
- anyviolence: our definition of any violence
- armedconf_dp : the number of months since the last armed conflict
- anyviolence_dp: the number of months since the last month with any violence

Which model should you use?

Our default model is the best model, armed conflict, 12 months ahead. This model predicts escalations of violence in with more than 5 fatalities per 1 million inhabitants of a country (using the latest World Bank population data for most countries). These are serious outbreaks of internal political violence for the respective country relative to their population. For countries like India or China this risk is lower due to their large population size.

Any violence is any fatality due to political armed conflict and here population does not play a role in the estimation of the risk. As the threshold here is lower, risk estimates are much higher for this type of conflict. The absolute threshold also makes more populous countries relatively riskier.

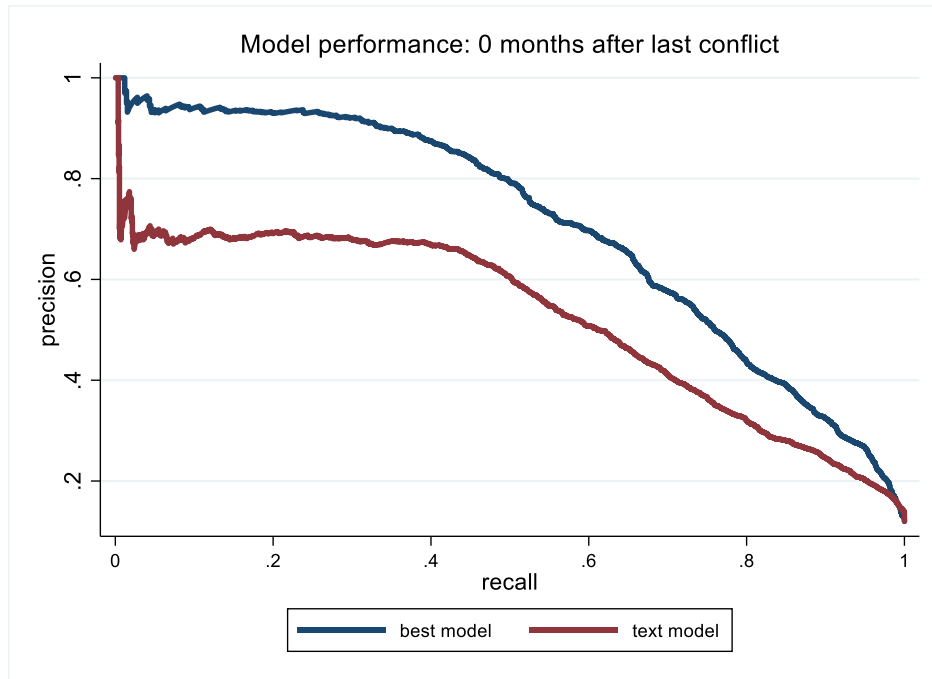
The 12 months ahead model uses a longer period to train the random forest. This means the model can pick up subtler signals but it also means that risks are more disperse. The 3 months ahead model will pick up more escalations of lower level violence like riots and activity by armed actors.

The question on which model to use therefore depends on the goal of using the forecast. For a measure of general political risks that might destabilize an entire country use the armed conflict forecast, 12 months ahead. For forecasts of violence itself use the any violence forecasts.

The best model is the better forecast overall as it uses conflict history in its forecast. However, this also means that the forecast is dominated by conflict dynamics and history. Countries which suffered recent, intense violence will be particularly risky. This means that some new developments will be forecasted slightly less precisely. The text model is trained with text alone to predict and it will, in many circumstances, even beat the best model in picking up new developments in a country.

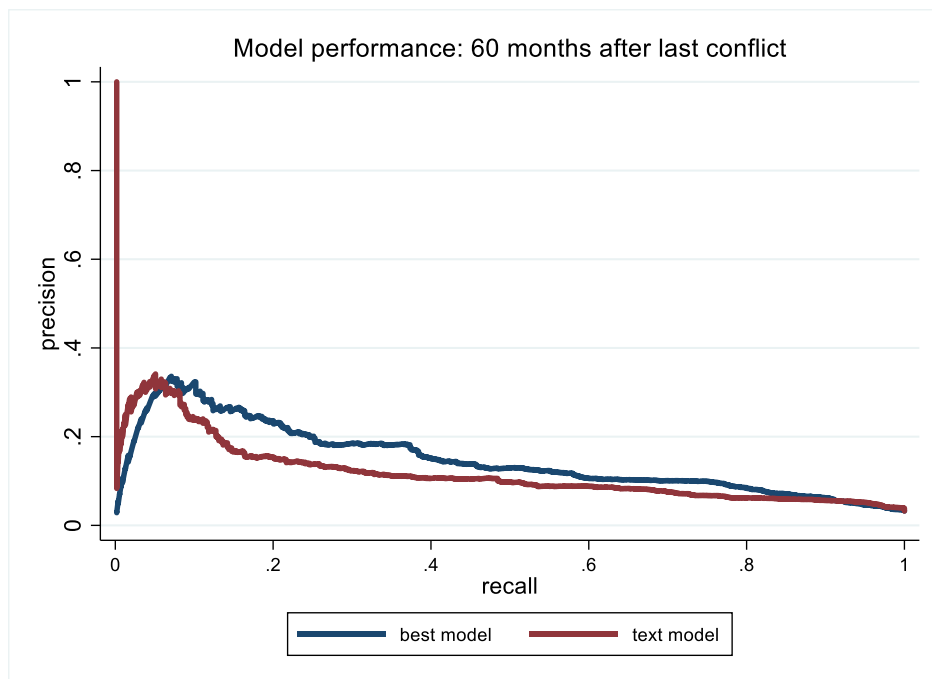
We recommend to never use the text model when predicting intensity. The best model performs much better in most circumstances in this model. We do not recommend the use of the intensity model in countries which have been at peace for several years.

The 12-ahead forecast has generally much higher precision even for lower AUC scores. It provides significantly less “false warnings”. A typical precision/recall curve for armed conflict onsets looks as follows:

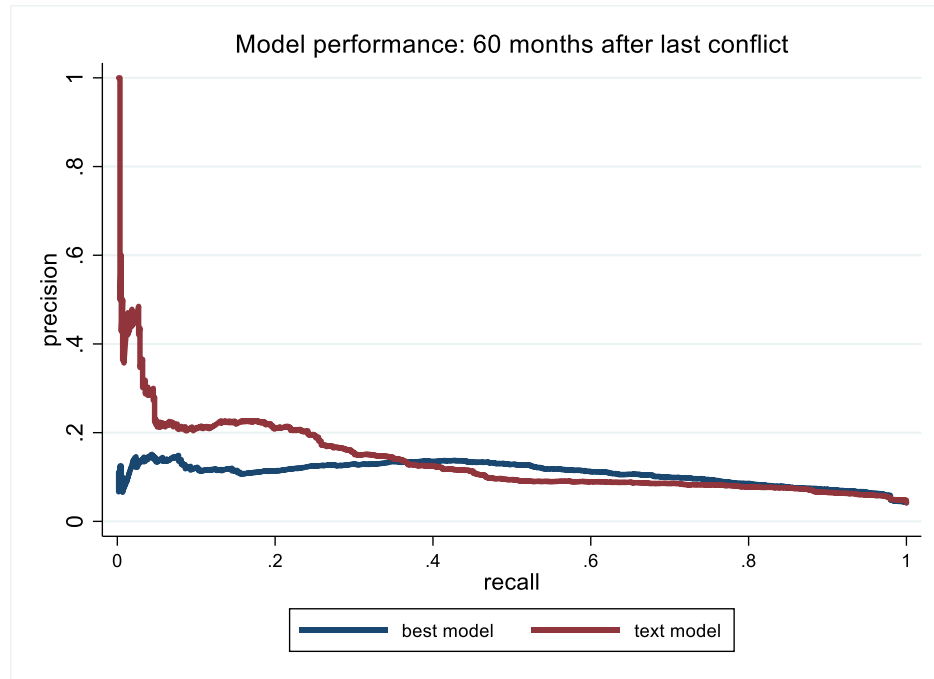


Precision/Recall curve armed conflict 12 months ahead, all onsets

This suggests extremely good performance of both models but the best/full model dominates consistently. However, when predicting hard-to-predict onsets this image changes somewhat. The typical precision/recall curve for armed conflict and any violence when predicting hard-to-predict onsets look as follows:



Precision/Recall curve armed conflict 12 months ahead, hard onsets



Precision/Recall curve any violence 12 months ahead, hard onsets

From this it is clear that the text model performs better and that armed conflicts are easier to predict. We almost reach a 20 % precision for 20% recall for these hard-to-predict onsets. For any violence performance is closer to 12% precision for 20% recall.

Please contact us if you need advice on how to use the forecasts or require specific data not provided in the download section.

Details on the method

Our method always uses information available by the end of a month to predict 3 or 12 months into the future. We collect the text information as quickly as possible after the end of the previous month. For the best model we need a valid conflict history which get the information around the 20th of the next month as this is the time at which the UCDP provides its estimates of fatalities. For example, on the 20th of February 2021 the UCDP released its estimates of fatalities during January 2021. We used this information, together with all text that has been published until the 31st of January, to predict conflict outbreaks after January.

When training on past data for performance measurement we do not use information on conflicts that have not broken out but will break out. For example, when predicting conflict from a standpoint of information in June 2020 we do not use the fact that we know a conflict broke out in July 2020. We do this by setting existing positive training observations to missing in the training sample when the conflict has not broken out yet. This means that the forecast in June 2020 cannot rely on the observation in May 2020 for that country. We also set observations to missing for which it is not clear, yet, whether a conflict

will break out. We are also not evaluating our model on these observations. For example, in May 2021 we will not evaluate performance on observations in the period April 2020 to May 2021 when forecasting 12 months ahead. We will not evaluate performance on observations in the period February 2021 to May 2021.

In our provision of risk statistics, we do not distinguish between countries which are in conflict from those that are not. And therefore countries with the highest risk estimates will mostly already suffer fatalities from armed conflict according to the UCDP.

For more technical details and background on the methodology please see: *Mueller, Hannes and Christopher Rauh (2022) The Hard Problem of Prediction for Conflict Prevention, Journal of the European Economic Association, 2022: <https://doi.org/10.1093/jeea/jvac025>*

Important Code Changes

8 May 2021: We spotted a minor coding error in our previous training and evaluation method – this made previous estimates and evaluations less valid for out-of-sample performance.

18 December 2022: We completely overhauled our method and switched text data source from LexisNexis to Factivia. We reinsured that forecasts are robust to this and performance is extremely similar in both the best and text model.

Performance stats in the armedconf12 model

Overall best model AUC 0.92 (old 0.91)

Overall text model AUC 0.87 (old 0.88)

Performance full model in the two months before conflict breaks out. AUC 0.96 (old 0.96)

Performance text model in the two months before conflict breaks out. AUC 0.92 (old 0.92)

Which means the forecast models become better as the conflict approaches.

Performance best model after 5 years of the previous conflict (harder to predict cases) AUC 0.82 (old 0.77)

Performance text model after 5 years of the previous conflict (harder to predict cases) AUC 0.78 (old 0.76)

Performance best model after 5 years of the previous conflict (harder to predict cases) in the two months before conflict breaks out. AUC 0.86 (old 0.81)

Performance text model after 5 years of the previous conflict (harder to predict cases) in the two months before conflict breaks out. AUC 0.81 (old 0.78)

Which means the forecast models retain some forecasting power in hard onsets.

Performance stats in the armedconf3 model

Overall best model AUC 0.94 (old 0.93)

Overall text model AUC 0.88 (old 0.89)

Performance full model in the two months before conflict breaks out. AUC 0.94 (old 0.94)

Performance text model in the two months before conflict breaks out. AUC 0.88 (old 0.90)

Which means the forecast models become better as the conflict approaches.

Performance best model after 5 years of the previous conflict (harder to predict cases) AUC 0.83 (old 0.76)

Performance text model after 5 years of the previous conflict (harder to predict cases) AUC 0.78 (old 0.76)

Performance best model after 5 years of the previous conflict (harder to predict cases) in the two months before conflict breaks out. AUC 0.84 (old 0.76)

Performance text model after 5 years of the previous conflict (harder to predict cases) in the two months before conflict breaks out. AUC 0.78 (old 0.76)

Which means the forecast models retain some forecasting power in hard onsets.

Performance stats in the anyviolence12 model

Overall best model AUC 0.89 (old 0.88)

Overall text model AUC 0.86 (old 0.84)

Performance full model in the two months before conflict breaks out. AUC 0.94 (old 0.94)

Performance text model in the two months before conflict breaks out. AUC 0.90 (old 0.90)

The models become better as the conflict approaches.

Performance best model after 5 years of the previous conflict (harder to predict cases) AUC 0.78 (old 0.79)

Performance text model after 5 years of the previous conflict (harder to predict cases) AUC 0.76 (old 0.72)

Performance best model after 5 years of the previous conflict (harder to predict cases) in the two months before conflict breaks out. AUC 0.80 (old 0.84)

Performance text model after 5 years of the previous conflict (harder to predict cases) in the two months before conflict breaks out. AUC 0.77 (old 0.74)

The models retain some forecasting power in hard onsets.

Performance stats in the anyviolence3 model

Overall best model AUC 0.92 (old 0.91)

Overall text model AUC 0.87 (old 0.87)

Performance full model in the two months before conflict breaks out. AUC 0.93 (old 0.92)

Performance text model in the two months before conflict breaks out. AUC 0.88 (old 0.88)

The models become better as the conflict approaches.

Performance best model after 5 years of the previous conflict (harder to predict cases) AUC 0.80 (old 0.77)

Performance text model after 5 years of the previous conflict (harder to predict cases) AUC 0.76 (old 0.72)

Performance best model after 5 years of the previous conflict (harder to predict cases) in the two months before conflict breaks out. AUC 0.81 (old 0.77)

Performance text model after 5 years of the previous conflict (harder to predict cases) in the two months before conflict breaks out. AUC 0.76 (old 0.72)

The models retain some forecasting power in hard onsets.